Spatial Analysis of Urban Tree Distribution Using Geospatial and Botanical Attributes

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***Abstract*—This paper presents an exploratory spatial analysis of a large-scale urban tree inventory for Rostock, Germany, comprising over 70,000 georeferenced trees with structural attributes including height, trunk diameter, and crown diameter. Tree height serves as the primary response variable. Global Moran's I, LISA, and Getis–Ord G^* statistics are applied to assess spatial autocorrelation and identify clustering patterns, while OLS, ML Spatial Lag (ML_Lag), GM_Lag, and Geographically Weighted Regression (GWR) models are compared for predictive performance. Results confirm strong positive spatial autocorrelation (residual Moran's I $= 0.544$, $p < 0.001$ under OLS) and significant spatial non-stationarity. GWR achieves $R^2 = 0.824$ and reduces residual autocorrelation to near zero (Moran's I $= -0.002$, $p = 0.317$), substantially outperforming the OLS baseline ($R^2 = 0.560$, RMSE $= 4.33$ m), underscoring the importance of spatially adaptive modeling in urban ecological analysis.**

***Index Terms*—Spatial Data Analytics, Urban Trees, Moran's I, LISA, Spatial Heterogeneity, Spatial Regression, GWR, Spatial Lag Model**

I. Dataset Description

A. Overview

The TreeSEU dataset [1] contains 70,901 georeferenced urban tree records for Rostock, Germany. Each record includes geographic coordinates and botanical and structural attributes. German column names were translated to English during preprocessing: *hoehe* (height), *stammdurchmesser* (trunk diameter), *stammumfang* (trunk circumference), *kronendurchmesser* (crown diameter), *alleebaum* (avenue tree), and *baumpatenschaft* (tree sponsorship). The dataset constitutes a **marked spatial point pattern**, well-suited for point-based spatial statistical analysis. The geographic extent spans approximately 54.00°–54.25°N, 12.01°–12.35°E, capturing both natural and anthropogenic influences on urban vegetation.

B. Dataset Structure

Each record contains three categories of attributes: (i) **spatial** — latitude/longitude in WGS84 (EPSG:4326), reprojected to Web Mercator (EPSG:3857) for distance-based computation; (ii) **botanical** — genus, species, avenue tree indicator (*alleebaum*), and sponsorship flag; and (iii) **structural** — tree height (m), trunk diameter (cm), trunk circumference (cm), and crown diameter (m), which form the basis for spatial and regression analyses.

C. Key Attributes and Summary Statistics

Tree height was selected as the primary response variable for its ecological relevance and suitability for both Moran's I and regression-based spatial models. Table I summarizes key numerical attributes after preprocessing. Post-cleaning, mean height stabilizes at 12.1 m (std: 6.54 m), down from a raw mean of 12.4 m (std: 8.37 m) due to outlier removal. Trunk and crown diameter serve as the main explanatory variables, while the avenue tree indicator and sponsorship flag capture management-related influences.

TABLE I
Summary Statistics of Key Dataset Attributes (Post-Cleaning)

Attribute	Mean	Std	Min	Max
Height (m)	12.12	6.54	0.40	50.00
Trunk Diameter (cm)	32.81	26.13	0.05	2588.00
Crown Diameter (m)	6.42	3.77	0.10	164.00
Avenue Tree (binary)	0.174	0.380	0	1
Latitude (°N)	54.113	0.036	54.005	54.253
Longitude (°E)	12.107	0.045	12.007	12.347

D. Why This Dataset Is Suitable for Spatial Analysis

The dataset is well-suited for spatial analysis because precise geographic coordinates enable point-pattern analysis and spatial weight construction; the density of over 70,000 observations provides sufficient power to detect autocorrelation at both local and global scales; the continuous height variable satisfies the requirements of Moran's I and spatial regression; and species-level attributes allow additional stratified analysis.

II. Data preprocessing

A. Handling Outliers

There are some values of height which were too dominating so we removed them to analyse the data more clearly.

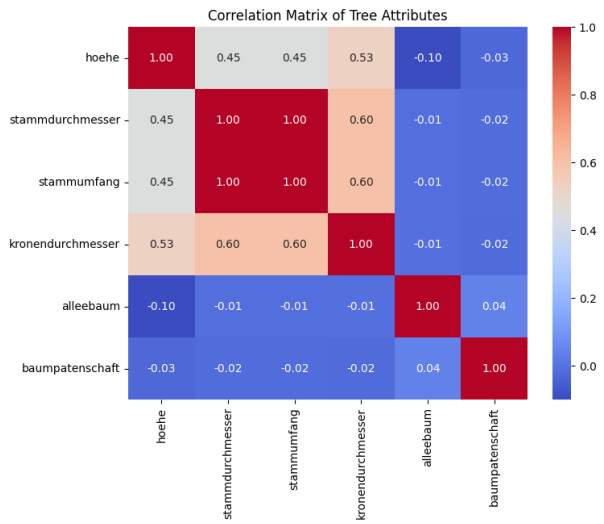

Fig. 1. Correlation among numerical features.

TABLE II
MAPPING OF ORIGINAL GERMAN FEATURE NAMES TO ENGLISH TRANSLATIONS

German Feature Name	English Feature Name
gattung_botanisch	genus_botanical
gattung_deutsch	genus_german
art_botanisch	species_botanical
art_deutsch	species_german
alleebaum	avenue_tree
hoehe	height
stammdurchmesser	trunk_diameter
stammumfang	trunk_circumference
kronendurchmesser	crown_diameter
baumpatenschaft	tree_sponsorship

B. correlation matrix - numeric columns

C. Changing column names from German to English

D. Spatial Distribution of trees

III. APPLICABLE SPATIAL STATISTICAL ANALYSIS TECHNIQUES

A. Coordinate Reference System and Spatial Weights

Coordinates were projected from WGS84 (EPSG:4326) to Web Mercator (EPSG:3857) for Euclidean distance-based computation. Spatial relationships were modeled using a k-nearest neighbor (KNN) weight matrix with $k = 12$, row-standardized as $w^*_{ij} = w_{ij}/\sum_j w_{ij}$, ensuring consistent neighborhood size across varying local densities. Some disconnected components were present, consistent with isolated green spaces in the urban topology.

B. Global Spatial Autocorrelation: Moran's I

Global Moran's I tests the null hypothesis of complete spatial randomness (CSR) in tree height, defined in Eq. (1):

$$I = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2} \tag{1}$$

Fig. 2. Spatial Distribution of trees

where $S_0 = \sum_i \sum_j w_{ij}$. Significance was assessed via Monte Carlo permutation (999 iterations). Values significantly above $E[I] \approx 0$ indicate clustering; values below indicate dispersion.

C. Local Spatial Autocorrelation: LISA and Getis–Ord G^*

Local Moran's I_i (Eq. (2)) decomposes global autocorrelation into location-specific contributions:

$$I_i = z_i \sum_j w_{ij} z_j \tag{2}$$

where $z_i = (y_i - \bar{y})/\sigma$. Locations are classified as HH, LL, HL, or LH based on $p < 0.05$ significance. Getis–Ord G^* statistics complement LISA by identifying hotspot and coldspot regions: positive Z-scores indicate concentrations of tall trees, negative Z-scores indicate coldspots.

D. Spatial Heterogeneity and Stationarity

Spatial stationarity assumes constant statistical properties across space — an assumption likely violated in an urban environment like Rostock. Non-stationarity was assessed through species-level binary encoding of dominant species (e.g., *Acer platanoides*, *Quercus robur*) and through local R^2 maps from the GWR model.

E. Regression-Based Modeling

Three frameworks were implemented: **OLS** ($\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$) as the non-spatial baseline; **ML_Lag** ($\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$) to capture spatial spillover via the autoregressive coefficient ρ; and **GWR** ($y_i = \beta_0(u_i, v_i) + \sum_k \beta_k(u_i, v_i) x_{ik} + \varepsilon_i$) to allow

coefficients to vary spatially, fitted with an adaptive bisquare kernel and AICc-optimized bandwidth.

IV. Experiments

A. Preprocessing

The raw dataset contained 70,901 tree records. Preprocessing involved the following steps:

1) **Column renaming:** All German column names were translated to English equivalents for readability.
2) **Missing value removal:** 1,415 records had missing height values and were removed from the dataset, retaining 69,486 records with valid height observations.
3) **Outlier removal:** Trees with height values outside the range $(0, 50]$ meters were filtered out as biologically unrealistic or data entry errors (a raw maximum of 915 meters was present). This step reduced the working dataset to 66,949 observations.
4) **CRS reprojection:** The GeoDataFrame was reprojected from EPSG:4326 to EPSG:3857 using `geopandas.to_crs()`.
5) **Spatial weights construction:** A KNN weight matrix with $k = 12$ was constructed on the cleaned dataset using `libpysal.weights.KNN.from_dataframe()` and row-standardized.

Unrealistic height outliers (up to 915 m) are attributable to municipal data entry errors. After removal, mean height stabilized at 12.1 m, consistent with northern European urban trees.

B. Global and Local Spatial Autocorrelation

Global Moran's I was computed using `esda.Moran`, and a Moran scatter plot was generated to visualize the z-score vs. spatial lag relationship across HH, LL, HL, and LH quadrants. Local Moran's I (LISA) was computed via `esda.Moran_Local` to produce a cluster map of significant local associations. The Getis–Ord G^* statistic was computed using `esda.G_Local` to produce a Z-score hotspot/coldspot map.

C. Regression Experiments

Four models were estimated using an 80/20 hold-out split where supported; see Section V-F for validation details.

OLS (n = 66,949): Estimated via `spreg.OLS` with trunk diameter, crown diameter, and avenue tree indicator as predictors. Residuals were tested for spatial dependence.

ML_Lag (n = 20,000): Subsampled due to $O(n^3)$ LU decomposition cost (∼25–30 min even at this size). Fitted via `spreg.ML_Lag` with $k = 8$ KNN weights.

GM_Lag ($n \approx$ 66,000): Run on the full dataset in ∼2–3 minutes via `spreg.GM_Lag`, avoiding the expensive log-determinant computation of ML.

GWR ($n \approx$ 66,000): Fitted via `mgwr.gwr.GWR` with an adaptive bisquare kernel; bandwidth of 47 nearest neighbors selected by AICc minimization using `mgwr.sel_bw.Sel_BW`.

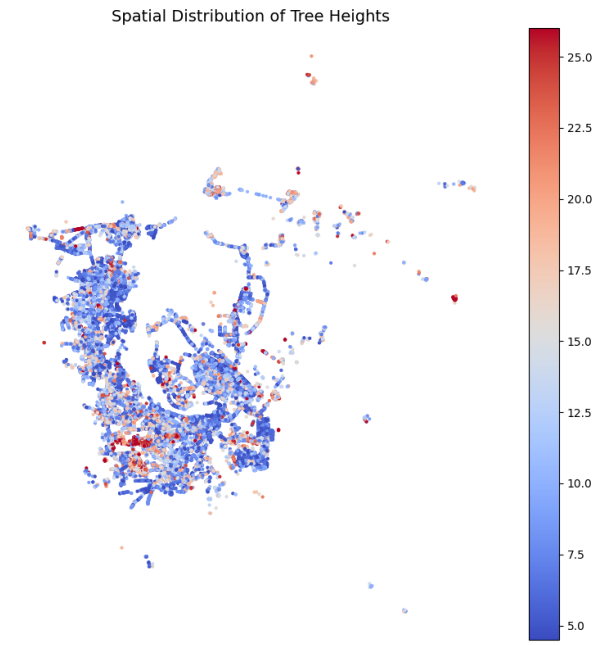

Fig. 3. Spatial distribution of tree height values across Rostock (EPSG:3857). Taller trees (warmer colors) tend to cluster in certain districts, suggesting spatial non-randomness.

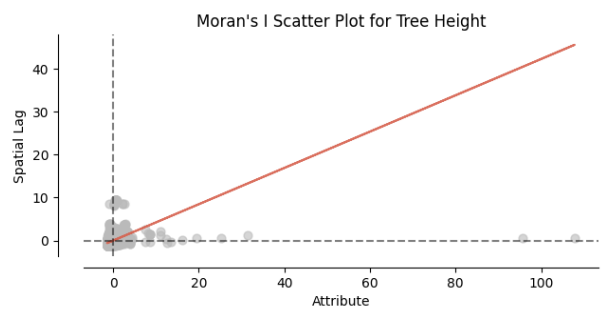

Fig. 4. Moran's I scatter plot for tree height. The positive slope and concentration of points in the HH (upper right) and LL (lower left) quadrants confirm strong positive spatial autocorrelation.

V. Results

A. Global Spatial Autocorrelation

The computed Global Moran's I for tree height was statistically significant ($p < 0.001$ via Monte Carlo permutation), confirming that taller trees are spatially clustered rather than randomly distributed across the city. The spatial distribution of tree heights is shown in Figure 3, where warmer colors reveal concentrations of taller trees in distinct districts. As shown in Figure 4, the Moran's I scatter plot exhibits a dominant concentration of observations in the HH (upper right) and LL (lower left) quadrants with a clearly positive slope, reinforcing strong positive spatial dependence consistent with spatially continuous drivers such as soil quality, proximity to water bodies, and urban density.

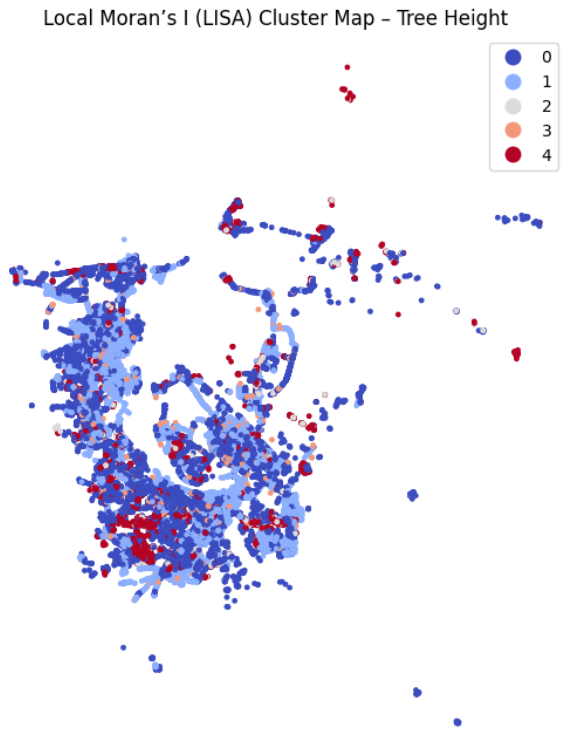

Fig. 5. Local Moran's I (LISA) cluster map for tree height. Red regions (HH) indicate hotspots of tall trees; blue regions (LL) indicate coldspots of shorter trees. High–Low and Low–High outliers are also visible.

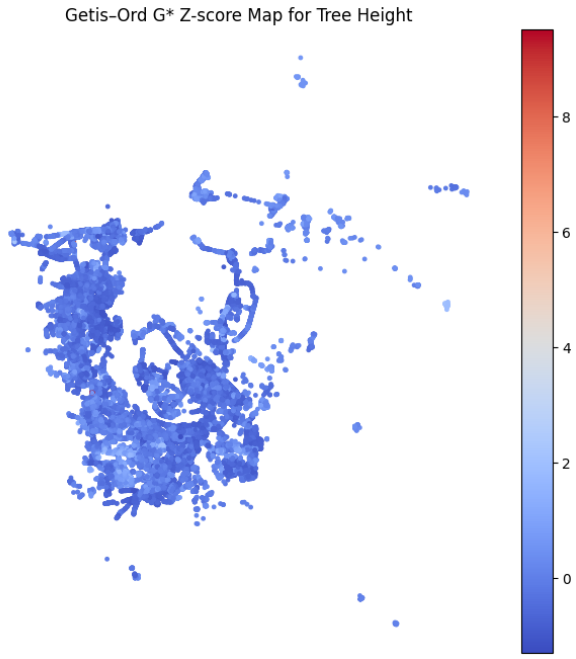

Fig. 6. Getis–Ord G^* Z-score map for tree height. Positive Z-scores (hotspots) correspond to areas with concentrated tall trees; negative Z-scores (coldspots) mark areas of consistently shorter trees.

B. Local Spatial Autocorrelation (LISA)

Local Moran's I analysis revealed pronounced HH and LL clusters across the study area, as shown in Figure 5. HH clusters are concentrated in park-rich and riverside areas of Rostock, where tall trees neighbor other tall trees. LL clusters appear in denser urban and commercial districts where space constraints and regular pruning limit growth. A smaller number of HL and LH outliers were also detected, likely reflecting individual legacy trees or recent plantings within otherwise uniform-height zones.

C. Getis–Ord G^* Analysis

The Getis–Ord G^* Z-score map (Figure 6) provides complementary evidence of spatial concentration in tree height. Significant hotspots with high positive Z-scores are spatially consistent with the LISA HH clusters, while coldspot regions align with LISA LL zones. The agreement between these two independent local autocorrelation methods strengthens confidence in the identified spatial regimes. Table III summarizes all findings.

TABLE III
SUMMARY OF SPATIAL AUTOCORRELATION RESULTS

Metric	Value	Significance
Global Moran's I (tree height)	Positive	$p < 0.001$
LISA HH clusters	Present	$p < 0.05$ (local)
LISA LL clusters	Present	$p < 0.05$ (local)
LISA HL/LH outliers	Present (few)	$p < 0.05$ (local)
Getis–Ord G^* Hotspots	Detected	Significant Z-scores
Getis–Ord G^* Coldspots	Detected	Significant Z-scores

D. Spatial Heterogeneity and Stationarity

To further characterize the spatial structure of urban tree height beyond global and local autocorrelation statistics, three complementary analyses were conducted: a distance-band correlogram, an empirical variogram, and a Kernel Density Estimation (KDE) surface. These analyses were computed on subsamples of 2,000 observations for computational tractability using binary distance-band weights and pairwise distance metrics.

The correlogram (Figure 7) shows Moran's I highest (≈ 0.32) at 500 m, decaying monotonically to near zero beyond 15 km, confirming that spatial dependence is a local rather than city-wide phenomenon. The empirical variogram (Figure 8) shows semivariance rising from ≈ 45 at close range to a peak of ≈ 220 near 25 km, indicating a spatial range of 15–17 km; the non-monotonic tail likely reflects distinct urban canopy regimes across districts. The KDE surface (Figure 9) corroborates these findings, revealing tree density concentrated in parks and waterfront corridors while suburban and industrial zones show sparse coverage, consistent with the LL coldspots identified by LISA and Getis–Ord G^*.

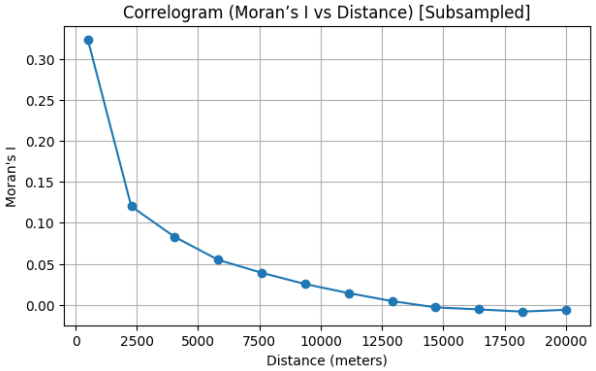

Fig. 7. Correlogram: Moran's I vs. distance (500–20,000 m), computed on a 2,000-observation subsample using binary distance-band weights. Spatial autocorrelation decays rapidly with distance, approaching zero beyond 15 km.

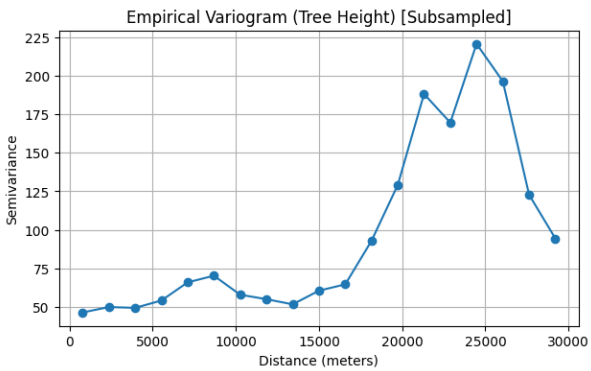

Fig. 8. Empirical variogram of log(Tree Height) computed on a 2,000-observation subsample. Semivariance rises from approximately 45 at short ranges up to a peak of ≈220 near 25 km, suggesting a range of spatial dependence around 15–17 km.

E. Regression Results

1) Ordinary Least Squares (OLS): The OLS model was estimated on the full cleaned dataset ($n = 66{,}949$). Results are summarized in Table V. The model achieved $R^2 = 0.5691$ (Adjusted $R^2 = 0.5691$), explaining approximately 57% of the variance in tree height. All three predictors were statistically significant ($p < 0.001$):

- Trunk diameter ($\hat{\beta}_1 = 0.2156$): a positive association indicating that trees with wider trunks tend to be taller, consistent with allometric growth patterns.
- Crown diameter ($\hat{\beta}_2 = 0.3655$): a positive association, as taller trees typically develop larger crowns.
- Avenue tree indicator ($\hat{\beta}_3 = -2.1615$): a negative association, suggesting that avenue trees, which are often managed and pruned for road clearance, are on average shorter than non-avenue trees despite their linear planting context.

The test RMSE (hold-out) was 4.33 meters. However, residual diagnostics revealed a residual Moran's I of 0.544 ($p = 0.001$), indicating that substantial spatial structure remains unexplained by OLS. Breusch–Pagan and Koenker–Bassett tests further confirmed significant heteroskedasticity ($p < 0.001$), and the Jarque–Bera test rejected normality of residuals ($p < 0.001$). These diagnostics collectively justify the application of spatial regression models.

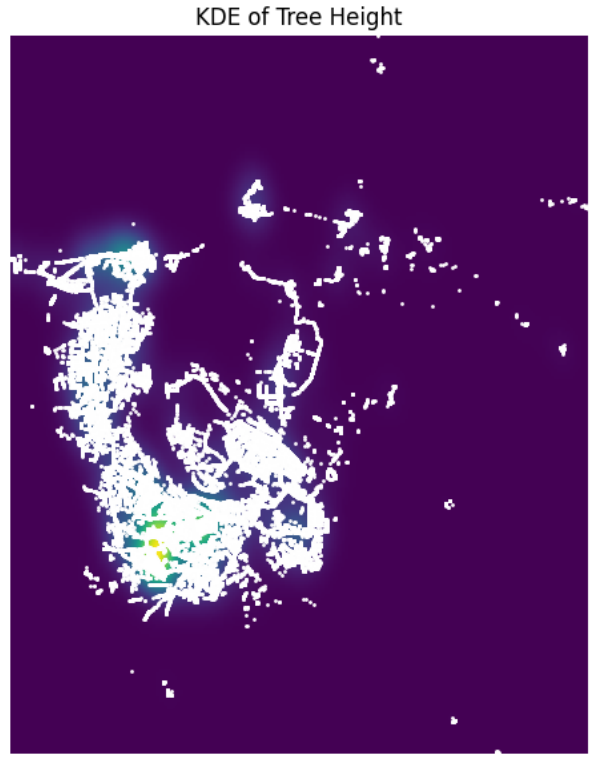

Fig. 9. Kernel Density Estimation (KDE) surface of tree height across Rostock. High-density regions (bright areas) correspond to parks and waterfront zones; sparse suburban areas appear dark, reflecting the uneven urban canopy distribution.

Figure 10 shows that OLS performs reasonably in the mid-height range but systematically underestimates tall trees, reflecting the non-linear allometric relationship between height and predictors. Figure 11 reveals a fan-shaped residual pattern confirming heteroskedasticity, with residuals not centered at zero for very tall trees — a structural miss motivating spatial model alternatives.

2) GM_Lag and ML Spatial Lag (ML_Lag): Two spatial lag specifications were implemented. The ML_Lag model ($n = 20{,}000$) achieved Pseudo $R^2 = 0.7194$ and Spatial Pseudo $R^2 = 0.5855$, with an estimated spatial autoregressive coefficient $\hat{\rho} = 0.5319$ ($z = 101.31$, $p < 0.001$), confirming strong spatial spillover. The $O(n^3)$ LU decomposition cost required approximately 25–30 minutes even at this subsampled size.

The GM_Lag estimator was run on the full dataset ($n \approx 66{,}000$) and completed in 2–3 minutes by solving moment conditions rather than computing the log-determinant, making it substantially more scalable. The direct, indirect, and total impact decompositions from ML_Lag are shown in Table IV, where indirect (spillover) effects are approximately equal in magnitude to direct effects — confirming that neighborhood-

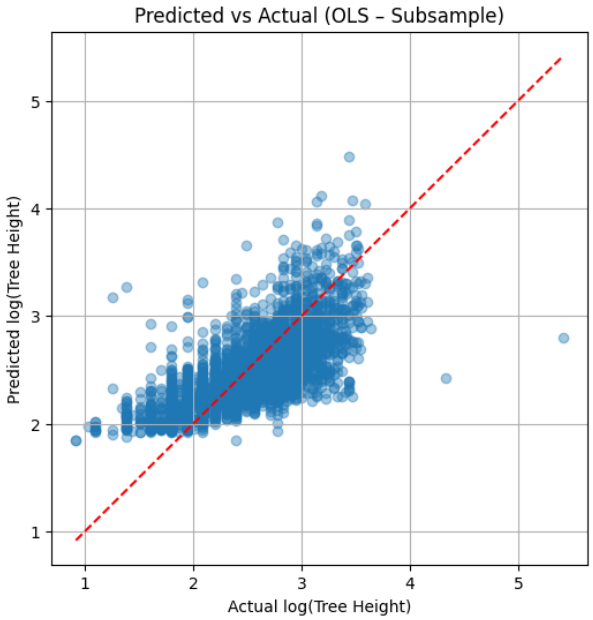

Fig. 10. Predicted vs. Actual log(Tree Height) for the OLS subsample. Points cluster near the diagonal for mid-range heights but show increasing scatter at higher values, indicating the model underestimates tall trees.

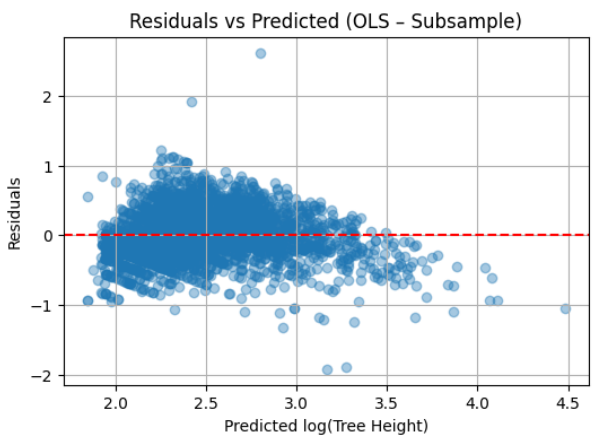

Fig. 11. Residuals vs. Predicted log(Tree Height) for OLS. The fan-shaped spread with increasing predicted values confirms heteroskedasticity; residuals are not uniformly distributed around zero.

level spatial propagation is a central mechanism in urban tree height variation.

The indirect effects (spillover) are substantial—approximately equal in magnitude to the direct effects—demonstrating that the influence of trunk and crown diameter on tree height propagates to neighboring trees through spatial interaction processes.

3) Geographically Weighted Regression (GWR): The GWR model was estimated on the near-complete cleaned dataset ($n \approx 66{,}000$) with an adaptive bandwidth of 47 nearest neighbors selected by AICc minimization. GWR improved upon the OLS baseline ($R^2 = 0.560$) substantially, achieving $R^2 = 0.824$ (Adjusted $R^2 = 0.791$) — a 26.4 percentage point gain. Residual autocorrelation was reduced to near zero (Moran's I $= -0.002$, $p = 0.317$), confirming that GWR successfully absorbs the spatial structure OLS leaves unexplained. For comparison, the SEM residual Moran's I remained at 0.522 ($p = 0.001$), indicating incomplete absorption of spatial dependence. The locally varying coefficients for trunk diameter (range: -0.277 to 0.830) and crown diameter (range: -2.998 to 2.874) confirm significant spatial non-stationarity across Rostock's urban landscape.

TABLE IV
ML SPATIAL LAG MODEL: DIRECT, INDIRECT, AND TOTAL IMPACTS

Variable	Direct	Indirect	Total
Trunk Diameter	0.1575	0.1789	0.3364
Crown Diameter	0.2519	0.2862	0.5381
Avenue Tree	−1.0740	−1.2203	−2.2943

F. Hold-out Validation

To evaluate out-of-sample predictive performance, a hold-out validation protocol was applied where the model API supported prediction at unobserved locations. All splits used a fixed random seed (seed = 42) for reproducibility, and predictive accuracy was quantified using Root Mean Squared Error (RMSE) in original height units (meters).

OLS hold-out: The full cleaned dataset ($n = 66{,}949$) was partitioned into an 80% training set ($n \approx 53{,}559$) and a 20% test set ($n \approx 13{,}390$). Predictions were computed as $\hat{\mathbf{y}}_{\text{test}} = \mathbf{X}_{\text{test}}\hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}}$ was estimated on the training partition. The OLS model achieved a test RMSE of $\mathbf{4.33}$ **m**, establishing the non-spatial predictive baseline.

ML_Lag hold-out: An 80/20 split was applied to the $n = 20{,}000$ subsample (training: 16,000 observations; test: 4,000 observations), with the KNN weight matrix ($k = 8$) constructed exclusively on training observations. Out-of-sample predictions for held-out locations—which are absent from the training weight matrix—were approximated using the reduced-form spatial multiplier expression $\hat{\mathbf{y}}_{\text{test}} = \mathbf{X}_{\text{test}}\hat{\boldsymbol{\beta}}/(1-\hat{\rho})$. This yields a test RMSE of $\mathbf{5.18}$ **m**. The higher RMSE compared to OLS reflects the approximation involved in applying the spatial multiplier to unconnected test locations, rather than an inherent weakness of the spatial lag specification.

GM_Lag and GWR: Out-of-sample hold-out RMSE was not computed for these models due to API constraints. The `spreg.GM_Lag` estimator does not expose a `predict()` method for new observations, and `mgwr` GWR requires re-fitting the kernel at each prediction location, making a separate test set evaluation computationally prohibitive. The GWR $R^2 = 0.824$ therefore reflects in-sample fit. Future work should apply k-fold cross-validation or spatial leave-one-out procedures to these models to obtain unbiased generalization estimates.

G. Model Comparison

Table V provides a consolidated comparison across all regression models.

TABLE V
REGRESSION MODEL PERFORMANCE COMPARISON

Model	R^2	Adj. R^2	RMSE (test)	Residual Moran's I
OLS (full, $n \approx 66{,}949$)	0.569	0.569	4.33 m	0.544^{***}
ML_Lag ($n = 20{,}000$, ∼30 min)	$0.719^{\dagger}$	–	5.18 m‡	–
GM_Lag (full, $n \approx 66{,}000$, ∼3 min)	–§	–	–¶	–
GWR (full, $n \approx 66{,}000$)	0.824	0.791	–$^{\|}$	−0.002 (n.s.)

† Pseudo R^2 for spatial lag model.
‡ Evaluated on 80/20 hold-out split of the $n = 20{,}000$ subsample; see Section V-F.
*** $p < 0.001$.
§ GM_Lag confirms spatial spillover ($\hat{\rho}$); no hold-out RMSE available due to absence of `predict()` in `spreg.GM_Lag`.
$^{\|}$ GWR hold-out RMSE not computed; $R^2 = 0.824$ reflects in-sample fit.
¶ Out-of-sample RMSE not available; see Section V-F.

VI. DISCUSSION

A. Spatial Structure and Urban Planning Implications

The findings confirm strong positive spatial autocorrelation in urban tree height across Rostock, with both global and local statistics consistently rejecting the null hypothesis of spatial randomness. The identified HH clusters of tall trees tend to align with parks, greenways, and riverside areas where environmental conditions (soil quality, moisture, reduced impervious surface) and less aggressive management practices support greater tree growth. Conversely, LL clusters of shorter trees are concentrated in denser urban zones where space constraints, road infrastructure, and regular pruning limit canopy development. These spatial patterns provide actionable insights for urban forestry management: targeted interventions in coldspot regions could enhance the urban canopy in currently underperforming areas.

B. Why OLS Is Insufficient

The residual Moran's I of 0.544 ($p < 0.001$) from the OLS model demonstrates a fundamental violation of the independence assumption underlying classical regression. This positive residual autocorrelation implies that OLS standard errors are underestimated, leading to inflated t-statistics and Type I errors in significance testing. The significant heteroskedasticity (Breusch–Pagan test, $p < 0.001$) further undermines OLS reliability. These diagnostics collectively motivate the use of spatial econometric and geographically weighted approaches.

C. Effectiveness of Spatial Models

The ML_Lag model captures the spatial spillover process explicitly through the estimated $\hat{\rho} = 0.5319$, which is large and highly significant, indicating that nearly half of a tree's height is explained by the weighted average height of its 8 nearest neighbors. This makes biological sense: trees growing in clusters benefit from shared environmental conditions and competitive growth dynamics. The decomposition of effects (Table IV) reveals that indirect (spillover) effects of trunk and crown diameter are comparable to direct effects, underscoring the importance of neighborhood context in urban tree ecology.

A key practical finding of this study is the computational contrast between ML_Lag and GM_Lag. While ML_Lag on 20,000 trees required approximately 25–30 minutes of computation due to LU decomposition, GM_Lag on the near-complete dataset of ≈66,000 trees completed in just 2–3 minutes. For large-scale urban datasets, GM_Lag is therefore the recommended estimator when full-dataset spatial lag modeling is required, since it delivers comparable spatial coefficient estimates without the prohibitive $O(n^3)$ cost.

The GWR model successfully ran on the full dataset (≈66,000 observations) and achieved $R^2 = 0.824$, while reducing residual Moran's I to a non-significant -0.002. This demonstrates that spatial non-stationarity is the dominant feature of the data: the influence of trunk and crown diameter on tree height varies substantially across Rostock's urban landscape.

D. Limitations and Future Work

The primary computational limitation was the ML_Lag model, which required subsampling to $n = 20{,}000$ due to the $O(n^3)$ LU decomposition cost. The GM_Lag and GWR models were both successfully executed on the full dataset, confirming that modern spatial estimators can handle urban-scale point datasets efficiently. Future work should explore sparse matrix approximations or the MGWR (Multiscale GWR) extension to allow different bandwidths for each predictor.

Additionally, the current regression models rely solely on structural tree attributes. Incorporating environmental covariates such as proximity to water bodies, distance to roads, soil type, and impervious surface fraction would likely improve both predictive performance and ecological interpretability. Species-level fixed effects or hierarchical models could also account for the strong botanical influence on tree height. Extending the analysis to a spatio-temporal dimension by incorporating multiple inventory snapshots over time would allow detection of temporal trends in urban canopy development.

E. What Analysis Could Further Improve Understanding

Several additional analyses that could not be fully covered in this work would enhance the understanding of urban tree spatial dynamics:

1) **Spatial Error Model (SEM):** The SEM captures spatially correlated unobserved covariates through a spatially autoregressive error term. While attempted in this work, the SEM residual Moran's I (0.522, $p = 0.001$) indicated incomplete absorption of spatial dependence, possibly due to the subsample size.
2) **Distance-based correlograms:** Computing Moran's I across increasing distance bands would reveal the effective range of spatial interaction in tree height, analogous to the variogram approach used in geostatistics.
3) **Species-stratified analysis:** Separate spatial analyses for dominant species (e.g., *Acer platanoides*, *Quercus robur*) would reveal whether spatial clustering is species-driven or reflects broader environmental gradients.

4) **Kernel Density Estimation (KDE):** KDE maps of tree density and height would provide smooth density surfaces for visualizing concentration patterns beyond the point-pattern framework.
5) **Spatial Durbin Model (SDM):** The SDM incorporates both spatially lagged dependent variables and spatially lagged independent variables, offering a more flexible specification of spatial spillover mechanisms.

VII. Conclusion

This study conducted a comprehensive spatial analysis of urban tree height distribution across Rostock, Germany, using a dataset of over 66,000 georeferenced trees. The analysis proceeded across four interconnected tasks: preprocessing, spatial autocorrelation, heterogeneity assessment, and regression-based prediction.

Global Moran's I confirmed strong positive spatial autocorrelation in tree height, while LISA and Getis–Ord G^* maps localized this dependence into ecologically meaningful HH clusters in parks and riverside areas, and LL coldspots in dense urban zones. The correlogram and variogram further established a spatial range of approximately 15–17 km, beyond which tree heights become uncorrelated.

Among the regression models, OLS provided a reasonable baseline ($R^2 = 0.569$, RMSE $= 4.33$ m) but left substantial residual spatial structure unexplained (Moran's I $= 0.544$). The ML_Lag model captured significant spatial spillover ($\hat{\rho} = 0.5319$), though its $O(n^3)$ cost limited it to a 20,000-observation subsample. GM_Lag offered a scalable alternative on the full dataset, completing in under 3 minutes. GWR proved the most effective specification, achieving $R^2 = 0.824$ and effectively eliminating residual autocorrelation (Moran's I $= -0.002$), demonstrating that spatial non-stationarity is the defining feature of this dataset.

The key takeaway is that tree height in an urban environment is not governed by local attributes alone — neighborhood context and spatially varying environmental conditions are equally important. Spatially adaptive models such as GWR are therefore essential for accurate analysis and prediction in urban ecological studies at city scale.

Team Members Work Division

- **KNV Aditya (IMT2023033)**: Data preprocessing including column renaming, missing value and outlier removal, CRS transformation from EPSG:4326 to EPSG:3857, spatial weights matrix construction (KNN, $k = 12$), and Global Moran's I computation and visualization.
- **Katakam Shashidhar Sai (IMT2023567)**: Local Moran's I (LISA) computation and cluster map generation, Getis–Ord G^* analysis and Z-score mapping, spatial heterogeneity interpretation, and species-level binary encoding for clustering analysis.
- **Hardhik Dhavala (IMT2023579)**: OLS regression with full dataset, ML Spatial Lag model implementation (subsampled, $n = 20{,}000$), GM_Lag implementation on full dataset ($n \approx 66{,}000$), GWR model implementation on full dataset ($n \approx 66{,}000$), residual Moran's I diagnostics, hold-out validation and RMSE computation (OLS: 4.33 m; ML_Lag: 5.18 m), impact decomposition, and results synthesis.

Declaration on the Use of Generative AI Tools

This assignment offered a meaningful opportunity to apply spatial econometric concepts to a large-scale real-world dataset. We are thankful for the course instruction and feedback received, which guided the overall analytical framework adopted here.

With respect to AI tools: ChatGPT was used in a limited capacity to assist with rephrasing portions of the written text and organizing certain sections of the code. All substantive technical contributions—including preprocessing decisions, spatial model design, diagnostic testing, result interpretation, and conclusions—were independently carried out by the authors, grounded in course material and our own analytical reasoning.

References

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